



BASIC LOGIC DESIGN AND SIMULATION

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A study and implementation of
basic logic circuits using
simulation tools



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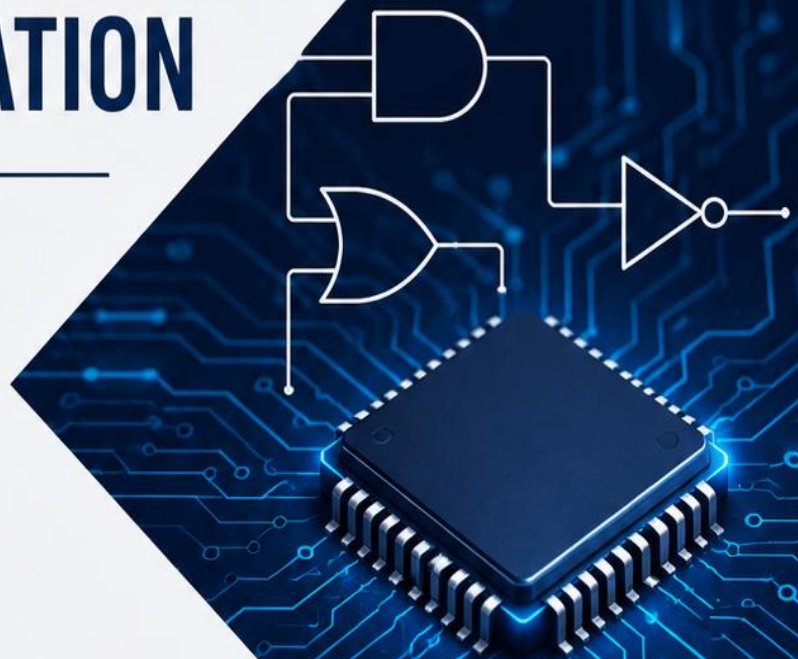
DESIGNATION:
VLSI Intern



ORGANIZATION:
CodTech IT Solutions



DATE:
24 May 2024



OVERVIEW

- Created and tested basic logic gates such as AND, OR, NOT, NAND, NOR, XOR, and XNOR using Tinkercad.
- Built the circuits with 74xx series ICs, switches, LEDs, and a power supply to understand how each gate works.
- Applied different combinations of binary inputs (0 and 1) and observed the corresponding outputs.
- Compared the obtained outputs with standard truth tables to verify the correctness of the circuits.
- Used Tinkercad Circuit Simulator to perform circuit testing without requiring physical hardware.
- Learned how Boolean expressions are related to the operation of digital logic circuits.
- Studied the behavior of different logic gates and analyzed their functions through simulation.
- Gained practical experience in designing, simulating, and testing digital electronic circuits.
- Identified and corrected circuit connection errors during the simulation process, improving troubleshooting skills.
- Developed a better understanding of the fundamental concepts used in digital electronics.
- Understood the role of logic gates as the basic building blocks of digital systems and computing devices.
- Successfully completed the simulation and verification of all logic gate circuits, obtaining results that matched the expected outputs.

OBJECTIVE

- To understand the working principles of basic logic gates.
- To design and simulate digital logic circuits.
- To verify gate functionality using truth tables.
- To gain hands-on experience with digital circuit simulation tools.
- To build a strong foundation for advanced VLSI and FPGA design concepts.

INTRODUCTION

Digital systems are built using logic gates, which perform simple logical operations on binary inputs. In this project, basic logic gates including AND, OR, NOT, NAND, NOR, XOR, and XNOR were designed and simulated to understand their working principles.

The circuits were created using Tinkercad Circuit Simulator along with 74xx series ICs, switches, LEDs, and a power supply. Different input combinations were applied to observe the output of each gate, and the results were compared with the corresponding truth tables to ensure correct operation.

Through this project, I gained practical exposure to digital circuit design and simulation. It also helped me understand how Boolean logic is implemented in electronic circuits and how individual logic gates can be combined to create more complex digital systems. The knowledge gained from this project provides a solid foundation for further study in areas such as VLSI Design, FPGA Development, ASIC Design, and Embedded Systems.

COMPONENTS USED

S.No.	Component	Quantity	Purpose
1	Breadboard	1	Platform for assembling the circuit
2	Power Supply (5V)	1	Provides power to the ICs and circuit
3	Input Switches/Push Buttons	2–4	Used to provide binary inputs (0 or 1)
4	LED Indicators	1–2 per circuit	Display the output state of the logic gate
5	Resistors (220Ω–330Ω)	As Required	Current limiting for LEDs
6	Connecting Wires	As Required	Electrical connections between components
7	IC 7408 (AND Gate)	1	Implements AND logic operation
8	IC 7432 (OR Gate)	1	Implements OR logic operation
9	IC 7404 (NOT Gate)	1	Implements NOT logic operation
10	IC 7400 (NAND Gate)	1	Implements NAND logic operation
11	IC 7402 (NOR Gate)	1	Implements NOR logic operation
12	IC 7486 (XOR Gate)	1	Implements XOR logic operation

S.No.	Component	Quantity	Purpose
13	Tinkercad Circuit Simulator	1	Used for circuit design and simulation

WORKING PRINCIPLE

The working of this project is based on the basic concepts of Boolean algebra and digital logic. Logic gates take binary inputs, either 0 or 1, and produce an output according to their specific logical operation. In this project, switches were used to provide input signals, while LEDs were used to display the output state of the circuit.

Different combinations of inputs were applied to each logic gate, and the corresponding outputs were observed through simulation. The obtained results were then compared with the standard truth tables to check whether the gates were functioning correctly. By performing these tests, it was possible to understand how each gate responds to different input conditions.

This project helped in understanding the practical implementation of digital logic and the role of logic gates in electronic systems. It also provided hands-on experience with circuit simulation, testing, and verification of gates such as AND, OR, NOT, NAND, NOR, XOR, and XNOR.

Working of Basic Logic Gates

1. AND Gate

Boolean Expression- $Y = A \cdot B$

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Working-An AND gate produces a HIGH output only when all its inputs are HIGH. If any input is LOW, the output remains LOW.

2. OR Gate

Boolean Expression- $Y = A + B$

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Working-An OR gate produces a HIGH output when at least one of its inputs is HIGH. The output is LOW only when all inputs are LOW.

3. NOT Gate

Boolean Expression- $Y = \bar{A}$

Truth Table

A	Y
0	1
1	0

Working-A NOT gate, also called an inverter, produces the opposite of the input signal. It converts 0 to 1 and 1 to 0.

4. NAND Gate

Boolean Expression- $Y = (A \cdot B)^{\bar{}}$

Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Working-A NAND gate is the inverse of an AND gate. It produces a LOW output only when all inputs are HIGH.

5. NOR Gate

Boolean Expression- $Y = (A + B)^{\bar{}}$

Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Working-A NOR gate is the inverse of an OR gate. It produces a HIGH output only when all inputs are LOW.

6. XOR Gate

$$\text{Boolean Expression-} Y = \bar{A}B + A\bar{B}$$

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Working-An XOR (Exclusive-OR) gate produces a HIGH output when the inputs are different. If both inputs are the same, the output is LOW.

7. XNOR Gate

$$\text{Boolean Expression-} Y = AB + \bar{A}\bar{B}$$

Truth Table

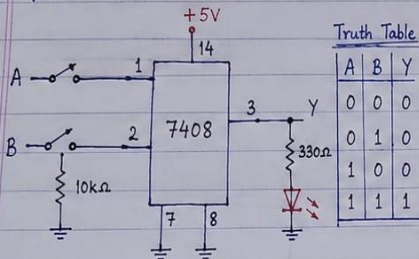
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Working-An XNOR (Exclusive-NOR) gate produces a HIGH output when both inputs are the same. It is the complement of the XOR gate.

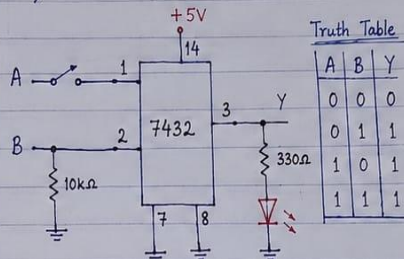
LOGIC GATE DESIGN AND TRUTH TABLE

BASIC LOGIC GATE DESIGN

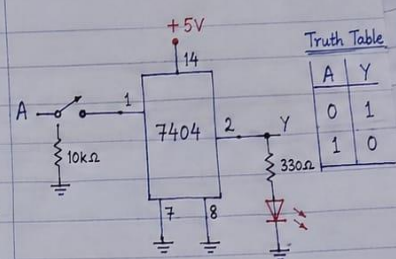
1) AND GATE (IC 7408)



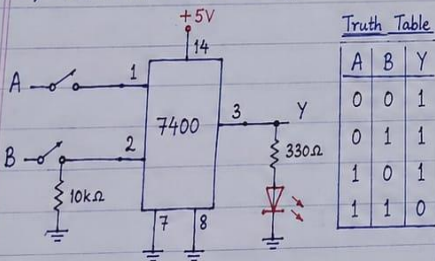
2) OR GATE (IC 7432)



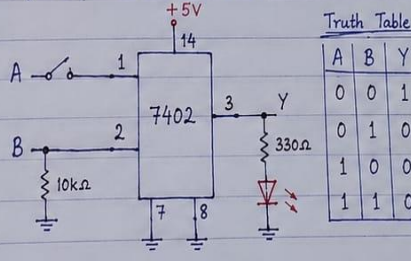
3) NOT GATE (IC 7404)



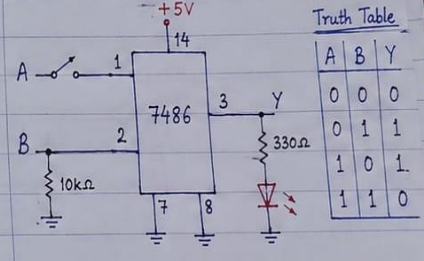
4) NAND GATE (IC 7400)



5) NOR GATE (IC 7402)



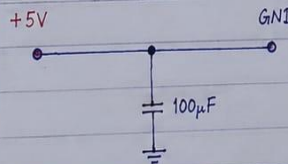
6) XOR GATE (IC 7486)



LEGEND

A, B → Inputs
 Y → Output
 10kΩ → Pull-up Resistor
 330Ω → Current Limiting Resistor
 LED → Output Indicator
 Switch → Input Switch

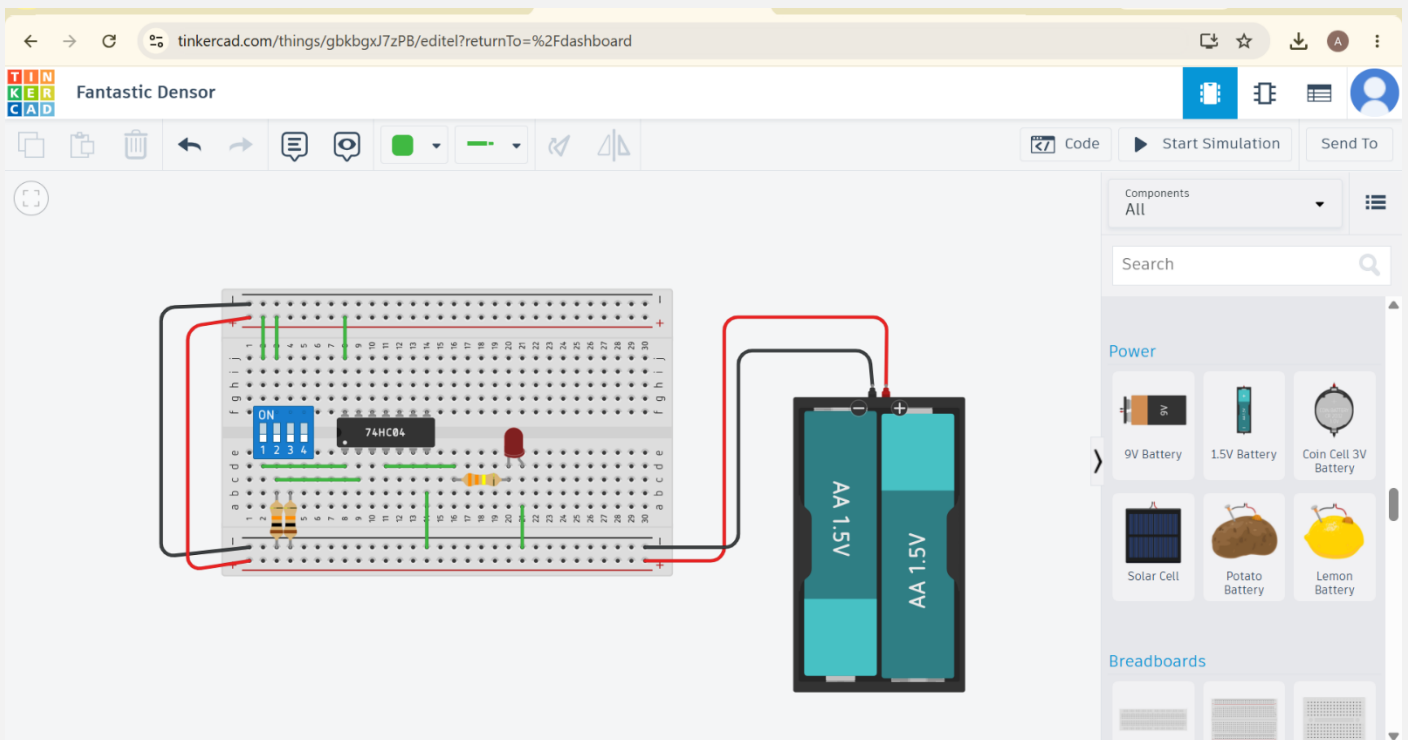
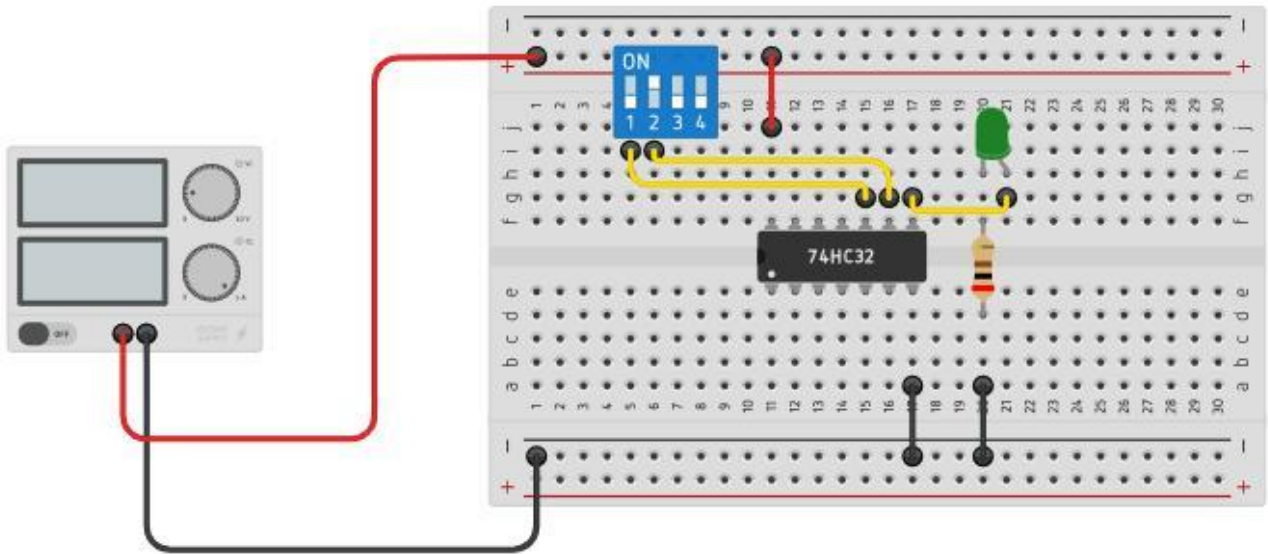
POWER SUPPLY



NOTES

1. All ICs are from 74xx series.
2. Pin 14 is connected to +5V (Vcc).
3. Pin 7 is connected to GND.
4. LED glows when output is HIGH (1).

SIMULATION IMAGES



METHODOLOGY SLIDE

Requirement Analysis



Circuit Design



Simulation in Tinkercad



Truth Table Verification



Result Analysis



Documentation

Conclusion

This project helped me understand the basic concepts of digital electronics by studying and simulating different logic gates. Through the design and testing of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates, I was able to observe how binary inputs are processed to generate the required outputs. The simulation results matched the expected truth tables, confirming the correct operation of each gate.

Working on this project improved my understanding of Boolean logic, digital circuit design, and circuit simulation. It also provided practical experience in using Tinkercad and implementing logic circuits using 74xx series ICs. Overall, this project strengthened my fundamentals in digital electronics and provided a good starting point for learning advanced topics such as VLSI Design, FPGA Development, ASIC Design, and Embedded Systems.

RESULT

- All gates simulated successfully.
- Outputs matched theoretical truth tables.
- No logical errors observed.
- Simulation validated digital logic principles.
- Project achieved all objectives.

CHALLENGES FACED

- Understanding IC pin configurations.
- Correct wiring in Tinkercad.
- Debugging incorrect output states.
- Verifying truth tables for multiple gates.

APPLICATIONS

- Arithmetic Logic Units (ALUs): Used to perform arithmetic and logical operations in processors.
- Microprocessors and Microcontrollers: Logic gates form the fundamental building blocks of CPUs and embedded systems.
- Digital Communication Systems: Used in data transmission, encoding, decoding, and signal processing circuits.
- Memory and Storage Devices: Essential for designing RAM, ROM, registers, and cache memory circuits.
- FPGA and ASIC Design: Logic gates are the foundation of programmable and custom integrated circuits.
- Control Systems: Used in automation, industrial controllers, and decision-making circuits.
- Embedded Systems: Implemented in smart devices, IoT applications, and consumer electronics.

Real-World Applications

- Traffic Light Controllers
- Digital Clocks
- Security and Alarm Systems
- Automatic Door Controllers
- Industrial Automation Systems
- Robotics and Smart Electronics

LEARNING OUTCOMES

Technical Skills

- Digital Logic Design
- Boolean Algebra
- Circuit Simulation
- IC Identification
- Digital Electronics Fundamentals

Professional Skills

- Documentation
- Technical Presentation
- Problem Solving
- Analytical Thinking

IMPROVE FUTURE SCOPE

Expand to:

- Half Adder Design
- Full Adder Design
- Multiplexer
- Demultiplexer
- Encoder
- Decoder
- ALU Design
- FPGA Implementation
- CMOS Layout Design
- VLSI Front-End Design

REFERENCES

- Digital Fundamentals by Thomas L. Floyd
- Morris Mano – Digital Design
- Tinkercad Documentation
- Texas Instruments 74xx Series Datasheets